rs1Wire: An Efficient Rust Ecosystem for 1-Wire Thermal Sensors

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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert.**

Introduction

INTRODUCTION INTRODUCTION

A. Subsection 1

More asdfasdfasdfasdfasdfasdfasdfasfasdfasdf details asdfasdfasdfasdfasdfasdfasdfasfasdfasdf about asdfasdfasdfasdfasdfasdfasdfasfasdfasdf Subsection asdfasdfasdfasdfasdfasdfasdfasfasdfasdf 1 asdfasdfasdfasdfasdfasdfasdfasfasdfasdf go asdfasdfasdfasdfasdfasdfasdfasfasdfasdf here asdfasdfasdfasdfasdfasdfasdfasfasdfasdf.

B. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

Related Work

Discuss related work ie: python modules and rust crates here. Make sure to explain what the differences and advantages/disadvantages of each are. Explain where the gap lies and what this crate does to address it.

Identify applicable funding agency here. If none, delete this.

C. Python

D. Rust

Background

Methodology

Show a figure of my hardware setup here
Also include the diagram of the circuit setup here.

Results

E. Figures and Tables

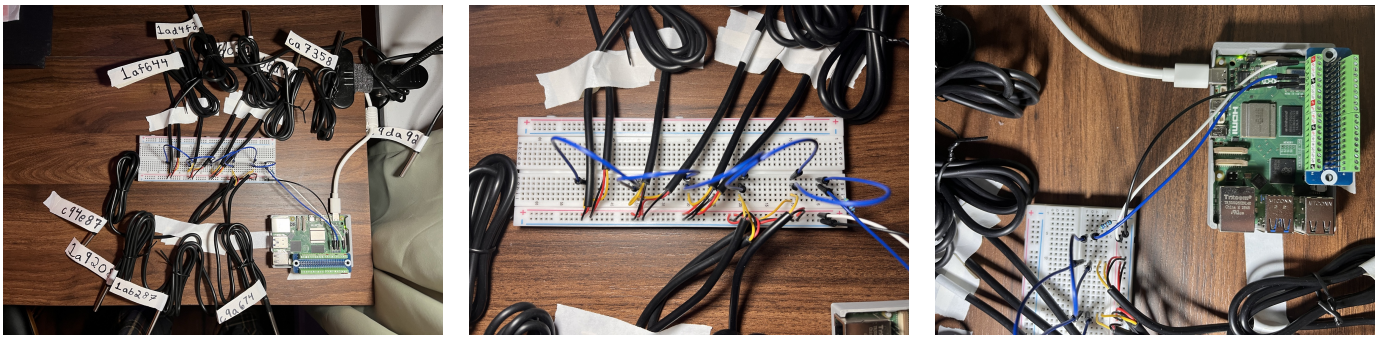
a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 3", even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
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[a]Sample of a Table footnote.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".



(a) Full setup (b) Breadboard (c) Raspberry Pi 5

Fig. 1. Test hardware: ten DS18B20 sensors on one w1-gpio bus.

Discussion

Conclusion

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(**??**)", not "Eq. (**??**)" or "equation (**??**)", except at the beginning of a sentence: "Equation (**??**) is . . ."

Acknowledgment

The author would like to thank all contributors and reviewers for their valuable feedback and support. Might acknowledge Professor Ashar here? Not sure if he gets his own spot in the author's list or just an acknowledgment as an advisor.

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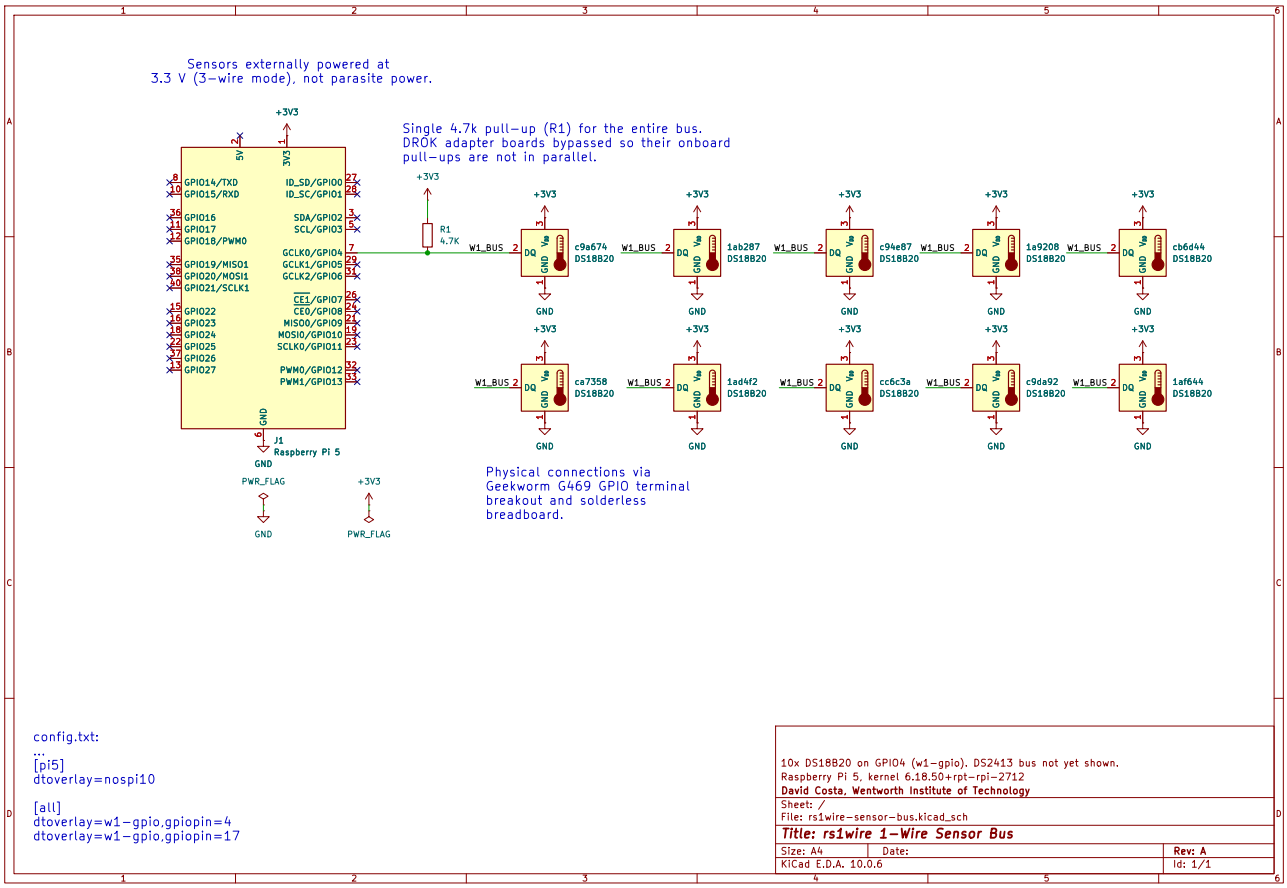


Fig. 2. Schematic of the 1-Wire sensor bus.

Fig. 3. Example of a figure caption.